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PROJECT CHAINWATCH: A USER CONFIGURABLE ANOMALY DETECTION SYSTEM FOR MONITORING BLOCKCHAIN NETWORK DATA FLOWS

Samson Qian, Steve Orrin, Matt Clanton and James Short

ABSTRACT

In this paper, we report on the motivation, design, testing, and implementation of a novel experimental system, ChainWatch, to watch and detect anomalies in blockchain network data flows using realistic blockchain network simulations and machine learning trained on data collected from those simulations. Although the machine learning systems are trained on a broad set of blockchain data and attacks, the test system itself is user-configurable and interpretable.

INTRODUCTION

Blockchain technology has the potential to revolutionize industries by providing a decentralized, secure, and transparent method for recording transactions. However, despite its robust architecture, blockchain networks are not immune to anomalies and malicious activities.[1] Robust anomaly detection in blockchain network data flows is crucial for maintaining the integrity, security, reliability, and trust in these systems.

Anomalies in blockchain networks can manifest themselves as unusual transaction patterns, deviations from normal network behavior, or unauthorized access attempts. Detecting these anomalies is essential for preventing fraud, ensuring compliance, and safeguarding user data. Traditional anomaly detection methods can fail due to the unique characteristics of blockchain, such as its decentralized nature and the immutability of its ledger.

In this paper, we report on the motivation, design, testing, and implementation of a novel experimental system, ChainWatch, to watch and detect anomalies in blockchain network data flows using realistic blockchain network simula tions and machine learning trained on data collected from those simulations. Although the machine learning systems are trained on a broad set of blockchain data and attacks, the test system itself is user-configurable and interpretable – that is, the user can select attack vectors from real and potential blockchain vulnerabilities, change attributes of the selected attack in real-time, and visualize how the attack corrupts the integrity and security of the blockchain. The trained anomaly detection models are deployed to this artificial blockchain network to continuously and proactively watch the blockchain network activity and node interaction to improve network security and performance and automate the anomaly detection process.

We organize our paper into four sections. First, we discuss selected background material and related work, and identify general design goals in developing a test environment for anomaly detection. Second, we discuss examples of anomaly scenarios that could be included in a test environment. Third, we discuss the specific design considerations for ChainWatch and present the test system architecture. We advise two external websites that provide further background information, example results, and the system test environment itself. Fourth, we comment on applicable use cases and directions for

[1] C. Ye, G. Li, H. Cai, Y. Gu, and A. Fukuda. Analysis of security in blockchain: Case study in 51%-attack detecting. In 2018 5th International Conference on Dependable Systems and Their Applications (DSA), pages 15–24, 2018.

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future research. Our work was carried out at BlockLAB, the blockchain research laboratory at the San Diego Supercomputer Center, UC San Diego, with research, systems, and administrative support from the Federal Division of Intel Corporation and the SPARK AI Industry Consortium.

BACKGROUND AND RELATED WORK

Blockchain was originally proposed as a secure technology to support cryptocurrencies. With the introduction of Ethereum in 2015, blockchain evolved with smart contracts, and computer programs that run on a blockchain in such a way that they can be executed and verified automatically. Each contract is developed as a sequence of operations whose cost is expressed as gas, or the amount of cryptocurrency necessary to execute the contract operations.[2] Despite its inherent trustless and decentralized design, Blockchain is not immune from attacks, however, encompassing a wide range of temporal attacks, spatial attacks, and logical-partitioning attacks. We give examples below of attack scenarios that have been attempted in recent years. A comprehensive review of different attack models, including using the concept of blockchain intelligence for anomaly detection or fraud detection in blockchain transaction data using AI, is reviewed in M Hasan, et. al. (2024).[3]

As with most systems, blockchain networks have inherent weaknesses that make them prone to third-party attacks. While the goal of blockchain is to create a trustless peer-to-peer network that provides scalability, decentralization, and security, often one of these is sacrificed in favor of the others. Most often, a network's security is sacrificed to promote greater decentralization, the goal of blockchain, and scalability, to facilitate a large pool of participants. For example, many networks may propose to significantly reduce the number of nodes to increase network throughput and facilitate more transactions. However, this makes the network vulnerable to many types of attacks if a party of nodes decides to act maliciously and take control of either a portion or the entire network.

Attacks can target either smart contract logic or the blockchain network itself. We focus on the latter as blockchain attacks arise not from code logic but from the design of the blockchain network. While there are other types of security risks and attacks in smart contract code, our research is particularly focused on potential network attacks on the underlying blockchain network, which is much more critical, as a new smart contract can easily be created and deployed on the chain but a vulnerable blockchain cannot easily be fixed and must be forked or replaced.

A prominent use case that has been theoretically explored is network detection and analysis of large-scale blockchain networks. Currently, blockchain cybersecurity attacks are predominantly handled reactively, often resulting in destructive techniques such as hard forks. As the importance of on-chain data intensifies and the value stored in smart contracts and digital wallets increases, the protection of blockchain networks becomes of greater importance. This network and blockchain activity monitoring research would be designed to simulate realistic blockchain network activity and collect relevant network and node data. The data created by this project can be used to explore the viability of automated blockchain network monitoring and attack detection through the use of machine learning and anomaly detection methods.

Our approach develops a user-configurable, visual analytical approach where node configuration, links, and statistics are collected and visualized through a chronological dashboard. We develop and train a machine learning model that can perform anomaly detection on blockchain network data and automatically detect if there is any malicious behavior occurring on the chain. Both the visual analytical tool and the machine

[2] F. Scicchitano, A. Lignori, M. Guarascio, E. Titacco, and G. Manco. A Deep Learning Approach for Detecting Security Attacks on Blockchain, ICAR-CNR, Rende, Italy

[3] M. Hasan, M.S. Rahman, H. Janicke, I.H. Sarker, Detecting Anomalies in Blockchain Transactions using Machine Learning Classifiers and Explainability Analysis.

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learning model make up ChainWatch, as the model is constantly watching all the activity on the blockchain to identify any anomalistic activity.[4]

DESIGNING A TEST ENVIRONMENT FOR ANOMALY DETECTION

Creating an effective test environment for anomaly detection in blockchain networks is a critical step in validating the efficacy of detection algorithms. The test environment should simulate real-world blockchain network conditions, including transaction volumes, network latency, and node distribution. It should also incorporate a variety of normal and anomalous data flows to ensure comprehensive testing.

Ideally, key components of an enterprise test system environment will include a sandboxed blockchain network, which allows for safe experimentation without affecting live systems, and a dataset that includes both typical transaction patterns and known anomalies. The environment should support the deployment of different anomaly detection models and provide tools for visualizing and analyzing their performance. Metrics such as detection accuracy, false positive rate, and processing time are essential for evaluating the effectiveness of the models under test.

Additionally, the test environment should be flexible enough to accommodate changes in network conditions and the introduction of new types of anomalies. This adaptability ensures that the detection system remains robust against evolving threats. By testing in a controlled environment, developers can refine detection techniques and enhance their reliability before deploying the detection system in a production blockchain network.

We considered several approaches to developing our blockchain test environment. For example, one could build and deploy an actual network of nodes to create a toy blockchain, and progress from that point. So, a second approach would be to collect relevant historical data of real large-scale blockchain networks, and attempt to replicate the distribution of that data to simulate realistic blockchain network activity. To facilitate both a useful and realistic representation of a live blockchain, we focused on designing and building our test system using the second approach.

BLOCKCHAIN NETWORK SIMULATION

Large-scale blockchain attacks are difficult to simulate precisely as there are many technical complexities with real large-scale attacks. To accurately simulate real large-scale blockchain networks and subsequent attacks, we build a simulation tool using historical blockchain data from large networks such as Bitcoin Core. This method allows for simulations of realistic and plausible blockchain network activity without requiring the significant amount of computing power of a real blockchain network. Another benefit of this framework is being able to easily simulate various types of blockchain network activity and attacks that may potentially provide information about those attacks. For example, transaction throughput, blockchain network latency, and node rewards can all be accurately simulated with an artificial blockchain sourcing from real historical network traffic data.

Furthermore, as the concepts of various network attacks may not be intuitive, we build a visual dashboard tool to simulate smaller-scale attacks that are easy to understand and follow. A partition attack on a blockchain network with hundreds of thousands of nodes may not be readily interpretable. However, the same attack on a smaller blockchain with only twenty or thirty nodes may make it much easier to understand what is happening behind the scenes. This has both the benefit of allowing users of ChainWatch to understand the underlying mechanics of network attacks, as well as being able to see the effectiveness of machine learning models in automatically detecting those attacks and anomalistic network behavior.

Monitored Metrics and Variables

To evaluate the performance and robustness of our anomaly detection system, we configured the test framework to monitor a targeted set of system-level and protocol-specific variables.

[4] F. Scicchitano ibid.

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These metrics were selected based on their relevance to both normal operational baselines and the manifestation of anomalous behavior. Specifically, we track throughput, throttle limits, and latency to assess network performance and responsiveness under varying load conditions. Area, transaction type distribution, and block/node rewards provide additional insight into spatial or role-based deviations in behavior, economic incentives, and usage patterns. Together, these variables form a comprehensive basis for identifying, characterizing, and responding to anomalies in real time. **Table One** summarizes the metrics and their purpose:

TABLE ONE: Metrics & Variables Monitored in Test System

Metric / Variable	Description	Purpose / Relevance
Throughput	Number of transactions processed per unit time	Measures network capacity and performance under load
Throttle	Rate-limiting mechanisms on transactions	Assesses system's ability to manage congestion and prevent overload
Latency	Time delay between transaction initiation and confirmation	Evaluates network responsiveness and real-time processing delays
Area	Spatial distribution or network topology aspects	Monitors how geographic or logical layout impacts behavior and anomalies
Transaction Type	Classification of transactions (e.g., transfers, contract calls)	Identifies patterns linked to specific transaction categories
Block and Node Rewards	Economic incentives given for block creation and validation	Analyzes incentive structures and their effect on node behavior

Experimental Design

Our test system architecture is designed to support flexible instrumentation across a range of experimental configurations, allowing us to isolate and evaluate the impact of key operational factors on anomaly detection performance. The following variables are central to our experimental design, as they influence both the behavior of the blockchain network and the complexity of detecting anomalous activity. These include protocol-level differences (e.g., variations in blockchain protocols and consensus algorithms), network composition (such as the number of nodes and the proportion of malicious participants), and system dynamics (including transaction volume per node, computation speed, and network bandwidth). By systematically varying these factors, we can assess detection robustness across diverse conditions and ensure that performance evaluations are generalizable across blockchain architectures and deployment scenarios. **Table Two** summarizes our experimental test factors and their objectives.

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TABLE TWO: Experimental Test Factors and Their Evaluation Objectives

Test Factor	Purpose of Variation	What It Helps Evaluate
Blockchain Protocol	Test across different protocol architectures (e.g., Ethereum, Hyperledger)	Generalizability of detection techniques across systems
Number of Nodes	Simulate small vs. large-scale networks	Scalability and behavior under network growth
Transactions per Node	Vary transaction volume per participant	Load sensitivity and throughput-related anomalies
Number of Malicious Nodes	Control adversarial presence in the network	Detection effectiveness, false positives under adversarial conditions
Computation Speed	Simulate nodes with varied processing power	System performance under heterogeneous hardware
Network Speed	Vary latency and bandwidth between nodes	Impact of network conditions on anomaly visibility and propagation
Consensus Algorithm	Evaluate with different consensus mechanisms (e.g., PoW, PoS, PBFT)	Sensitivity to consensus-specific vulnerabilities and attack vectors

ANOMALY DETECTION METHODS FOR DETECTING ATTACKS ON NETWORK DATA

The most common types of blockchain attacks typically lead to similar outcomes, only differentiated by the scale of the attack and the magnitude of the impact. For example, a partition attack typically results in a single node or group of nodes that have a consistently greater distribution of rewards than the other nodes it is partitioning out. The magnitude of this difference depends on the scale of the attack and the size of the blockchain network. However, the general idea is the same for all blockchain network sizes.

These patterns can be learned and detected by machine and deep learning methods. By training these models on large amounts of simulated blockchain network data, the models can learn deep complex patterns of what a blockchain attack can look like and how it impacts the regular flow of network activity. For example, a deep learning model can quickly spot and identify irregularities or anomalies in network latency, transaction throughput, node reward distributions, or a combination of some or all of these factors. This can be done through either a supervised or unsupervised approach, in which we will explore the uses, benefits, and weaknesses of both.

With the previously mentioned simulation framework, large amounts of simulated blockchain network data can be generated to train the model. Furthermore, malicious activity and malicious nodes can easily be integrated into the network to allow the model to learn what malicious activity looks like and how it impacts or deviates from the flow of network activity.

A Summary of Machine Learning Methods

Autoencoders: Unsupervised approach trained with only data on normal/typical blockchain network activity (no attacks/anomalies) to model the distribution of normal/typical blockchain

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activity. Then, when irregular or atypical network activity appears that deviates from the learned distribution of normal activity, the model will identify and flag these instances as anomalistic and potential attacks.

Sequential Models/LSTMs: Supervised approach trained with both normal/typical blockchain network activity and attacks to directly model and learn what factors contribute most to identifying blockchain attacks. Model learns a probabilistic distribution and outputs a binary classification at each blockchain activity update of whether the activity displays anomalistic/attack behavior, or not.

Graph Neural Networks: Using deep learning methods combined with learning a graph representation of the blockchain network, based on which nodes are connected in the network. A graph representation, which other models lack, provides additional information and context about which nodes are interacting with each other. This can help deep learning models detect attacks such as partitioning attacks with more ease if the model knows which nodes are isolated within a network.

Ensemble Methods: A combination of the above methods, employing multiple models using various techniques and learning methods to make predictions, then aggregating the results/outputs to make one conclusion. The idea is that we gain more confidence in predictions and reduce the likelihood of false positives and false negatives by having multiple models monitoring the blockchain simultaneously.

EXAMPLES OF BLOCKCHAIN ANOMALY SCENARIOS

As noted, typically the most common types of blockchain attacks are only differentiated by the scale and magnitude of the attack. Below are examples of anomaly scenarios that can be instrumented and tested in our environment:

Unusual Transaction Patterns:

- High-Frequency Transactions: A single address suddenly starts making a large number of transactions in a short period.
- Large Transactions: An address that typically makes small transactions suddenly initiates a very large transaction.
- Micro-Transactions: Numerous small transactions that are not typical for the network, potentially indicating a dusting attack.

Deviations from Normal Network Behavior:

- Sudden Spike in Network Traffic: A significant increase in the number of transactions or data flow within a short time frame.
- Unusual Node Activity: A node that was previously inactive becomes highly active, or a normally active node suddenly goes offline.
- Latency Anomalies: Unexpected delays in transaction processing times, which could indicate network congestion or a potential attack.
- Reward Distribution: A consistent and large imbalance of reward distribution between certain nodes (i.e. if a single node or group of nodes are consistently receiving greater rewards than other nodes)

Unauthorized Access Attempts:

- Failed Login Attempts: Multiple failed attempts to access a node or wallet, suggesting a brute force attack.
- Unauthorized Transactions: Transactions initiated without proper authorization, possibly indicating a compromised private key.
- Access from Unusual Locations: Nodes or wallets being accessed from IP addresses or geographic locations that are not typical for the network.

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Smart Contract Anomalies:

- Unexpected Contract Calls: Smart contracts being called in ways that are not consistent with their intended use.
- Reentrancy Attacks: Exploits where a smart contract is repeatedly called before the previous execution is completed, leading to potential double-spending.

Consensus Mechanism Anomalies:

- Forking Events: Unexpected forks in the blockchain could indicate an attack on the consensus mechanism.

Double-Spending Attempts:

- Transactions that attempt to spend the same cryptocurrency more than once.
- Repeated scenario testing will provide the data to evaluate the robustness and effectiveness of the test environment in identifying and mitigating potential threats.

CURRENT AND FUTURE USE CASE APPLICATIONS FOR CHAINWATCH

As enterprise adoption of blockchain technology, which had slowed in recent years, begins to regain momentum, ensuring the integrity and security of these systems is becoming increasingly important.[5] One powerful tool in this effort is blockchain anomaly detection, which helps identify irregular or suspicious patterns in data that may signal fraud, system failures, or security breaches. Currently, these detectors play a critical role in financial fraud prevention, supply chain integrity, healthcare data security, smart contract reliability, and IoT device monitoring. Looking ahead, their importance is expected to grow in areas such as decentralized governance, secure digital voting, energy trading, identity verification, and regulatory compliance. These applications highlight how anomaly detection can help build trust, reduce risk, and enable safe and efficient use of blockchain technologies.

[5] Dashveenjit Kaur, Enterprise blockchain: The hype is over – now comes the real work. The Block, April 28,2025. Available at: https://blockchaintechnology-news.com/news/enterprise-blockchain-the-hype-is-over-now-comes-the-real-work/

Examples of Current Use Cases

Fraud Detection API in Financial Transactions: Blockchain anomaly detectors are used to identify suspicious transactions and prevent fraud in financial systems. For example, detecting unusual patterns in cryptocurrency transactions can help prevent money laundering and other illicit activities.[6]

Supply Chain Monitoring; Anomaly detection can be used to ensure the integrity of supply chains by identifying discrepancies in the recorded data, such as unexpected delays or unauthorized changes in shipment details.[7]

Healthcare Data Security: Blockchain anomaly detectors help secure sensitive healthcare data by identifying and mitigating potential breaches or unauthorized access to patient records.[8]

Smart Contracts: Ensuring the proper execution of smart contracts by detecting anomalies in the contract execution process can prevent exploitation and ensure compliance with predefined conditions.[9]

[6] Mohammad Hasan, Detecting Anomalies in Blockchain Transactions using Machine Learning Classifiers and Explainability Analysis, arXiv:2401.03530v1 [cs.LG] 7 Jan 2024.

[7] Liu, Z., Gao, H et. al. (2024). Blockchain Anomaly Transaction Detection: An Overview, Challenges, and Open Issues. In: Qiu, X., Xiao, Y., Wu, Z., Zhang, Y., Tian, Y., Liu, B. (eds) The 7th International Conference on Information Science, Communication and Computing. ISCC2023 2023. Smart Innovation, Systems and Technologies, vol 350. Springer, Singapore. https://doi.org/10.1007/978-981-99-7161-9_10

[8] World Economic Forum, How Blockchain Can Improve Data Security in Healthcare (2023). https://www.weforum.org/stories/2023/12/healthcare-data-breaches-blockchain-cybersecurity/

[9] Muneeb Ul Hassan, Mubashir Husain Rehmani, and Jinjun Chen (2021). Anomaly Detection in Blockchain Networks: A Comprehensive Survey. arXiv:2112.06089v1 [cs.CR] 11 Dec 2021

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Prospective Future Use Cases

Decentralized Autonomous Organizations (DAOs): Anomaly detection can be used to monitor and secure the operations of DAOs, ensuring that decisions and transactions are legitimate and free from manipulation.[10]

Voting Systems: Blockchain anomaly detectors can be employed to secure electronic voting systems, detecting and preventing tampering or fraudulent activities during elections.[11]

Energy Trading: In decentralized energy trading platforms, anomaly detection can help identify and prevent fraudulent transactions and ensure the integrity of energy trading data.[12]

Digital Identity Verification: Blockchain anomaly detectors can enhance the security of digital identity verification systems by identifying and preventing identity theft and fraud.[13]

Regulatory Compliance: Future blockchain anomaly detectors can assist organizations in ensuring compliance with regulatory requirements by detecting and reporting any deviations from established rules and standards.[14]

CONCLUDING REMARKS AND FUTURE BLOCKCHAIN SECURITY RESEARCH

Typically, it requires considerable effort to identify the root causes of blockchain attacks, although they may share similar symptoms and outcomes. Machine learning and anomaly detection methods can help identify and understand the most important factors, or combination of factors, that suggest an attack, or potential attack.

Improving the security and performance of blockchain networks is of increasing importance as blockchain technology continues to advance. Today, there is no standard for monitoring blockchain attacks and suspicious activity because of the complexity of identifying and collecting relevant data for predicting attacks. Attacks are often spotted long after they occur and lead to irreparable damage to the blockchain network.

Chainwatch uses the simulation of real-world blockchain networks that allows for data collection in association with different events that can happen on a blockchain network. Aggregating pre-validated off-chain data associated with specific blockchain network activity will offer insight and solutions into ways in which blockchain networks can improve their security and performance. This is done through machine learning-enabled anomaly detection models that

[10] Michael Zargham and Kelsie Nabben (2023), Aligning 'Decentralized Autonomous Organization' to Precedents in Cybernetics. Available at: https://law.mit.edu/pub/dao-precedents-cybernetics/release/1

[11] Said El Kafhali(2023). Blockchain-Based Electronic Voting System: Significance and Requirements. https://onlinelibrary.wiley.com/doi/10.1155/2024/5591147

[12] G. B. Bhavana, R. Anand et. al. (2024). Applications of blockchain technology in peer-to-peer energy markets and green hydrogen supply chains: a topical review. https://www.nature.com/articles/s41598-024-72642-2

[13] Yang Liu and Debiao He et. al.(2020). Blockchain-based identity management systems: A review. Journal of Network and Computer Applications, Volume 166, 15 September 2020, 102731. Also see Andrej J. Zwitter et. al. (2020). Digital Identity and the Blockchain: Universal Identity Management and the Concept of the "Self-Sovereign" Individual. Frontiers in Blockchain. https://www.frontiersin.org/journals/blockchain/articles/10.3389/fbloc.2020.00026/full

[14] Primavera De Filippi and Samer Hassan (2018). Blockchain Technology as a Regulatory Technology: From Code is Law to Law is Code. arXiv:1801.02507 [cs.CY]. Also see Miriam Allena (2020). Blockchain Technology for Environmental Compliance. Environmental Law. Vol. 50, No. 4 (2020), pp. 1055-1103.

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learn from attack simulations and analyze the common patterns that appear.

This proof-of-concept framework can be adapted and utilized in large-scale blockchain projects, whether for blockchain maintainers or regulators, to monitor suspicious activity and potential attacks on the network. The Chainwatch API uses its pre-trained machine learning anomaly detection model to connect to a particular blockchain and continuously monitor and flags any potential suspicious activity or attacks. We hope this framework will be useful to blockchain developers to continuously test and improve the robustness of their blockchain network. And for regulators to quickly flag and identify malicious participants within the network.

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Disclaimers

The research presented in this brief reflects the authors' independent work and is not intended to represent the views of any organization or individual affiliated with the project.

About the Authors

Samson Qian is an independent researcher and member of the SPARK AI Consortium Advisors. Samson led the design and programming of the demonstration system.

Steve Orrin is the Chief Technology Officer for the Federal Division of Intel Corporation.

Matt Clanton is a member of the SDSC Operations Group and served as the project manager.

James Short is Lead Scientist and Director of the SPARK AI Industry Consortium.

Requesting Access to the Demo System

Access to the project's demonstration site is subject to approval by the authors. All requests should be directed to the corresponding author.

Corresponding Author

Direct all correspondence to James Short at jshort@ucsd.edu

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APPENDIX A: EXAMPLE PAGES FROM THE SPARK AI BLOCKLAB DETECTOR

The following pages are abstracted from the BlockLab ChainWatch detector website for purposes of illustrating the detector dashboard. As this is an active research project, the website is in development. Interested parties may contact the corresponding author for further details.

Concept and Design

- To design a novel experimental system, Chainwatch, to watch and detect anomalies in blockchain network data flows using realistic blockchain network simulations and machine learning trained on data collected from those simulations.
- Although the machine learning systems are trained on a broad set of blockchain data and attacks, the test system itself is user-configurable and interpretable – that is, the user can select attack vectors from real and potential blockchain vulnerabilities, change attributes of the selected attack in real-time, and visualize how the attack corrupts the integrity and security of the blockchain.
- The trained anomaly detection models are deployed to this artificial blockchain network to continuously and proactively watch the blockchain network activity and node interaction to improve network security and performance, and to automate the anomaly detection process.

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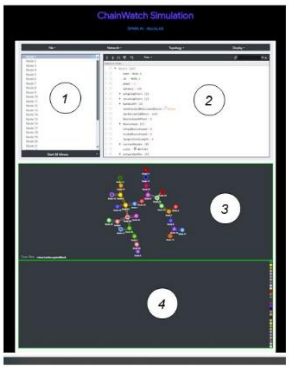
Test System Metrics Instrumented

Our test system architecture is designed to instrument and test:

1. Metrics & Variables to Monitor
   i. Throughput
   ii. Throttle
   iii. Latency
   iv. Area
   v. Transaction Type
   vi. Block and Node Rewards
2. Factors to Test in Different Experimental Designs
   i. Blockchain Protocol
   ii. Number of Nodes
   iii. Number of Transactions/Nodes
   iv. Number of Malicious Nodes
   v. Computation Speed
   vi. Network Speed
   vii. Consensus Algorithm

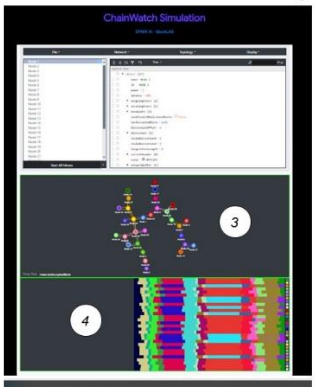
Screen Shot of ChainWatch Dashboard



Visual Dashboard

1. Nodes
2. Node Structure Information
3. Node Display
4. Visualization Window

ChainWatch Running: What is Displayed



Visual Dashboard

1. Nodes Configured
2. Node Structure Configured
3. Node Display Shows Node Transactions (colors iterate)
4. Visualization Window Shows Cumulative View of Node Transactions Running in Real Time

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APPENDIX B: A HYPOTHETICAL TEST SEQUENCE FOCUSED ON DETECTING UNUSUAL NODE BEHAVIOR

The following is an example test sequence designed to simulate and detect a variety of normal and malicious node activities within a controlled environment. The test begins by initializing a simulated network topology with a representative distribution of honest nodes, each configured to perform standard protocol operations such as block propagation, transaction validation, and consensus participation. Baseline metrics are collected to define normal behavioral patterns, including message frequency, peer connectivity, latency, and resource usage.

Subsequently, the environment introduces anomalous node behavior in a staged manner. This includes launching Sybil attacks by injecting multiple pseudonymous nodes attempting to disrupt consensus or flood the peer discovery protocol. In another phase, a subset of nodes may exhibit behaviors consistent with denial-of-service (DoS) attacks, such as selectively dropping messages, propagating malformed blocks, or artificially delaying transaction relay. Additional scenarios may involve compromised nodes executing eclipse attacks, isolating target nodes from the rest of the network. Throughout the test, the detection system is assessed on its ability to flag deviations in communication patterns, peer dynamics, and node reliability in real time.

By systematically introducing a spectrum of node-based threats under varying network conditions—such as increased latency, reduced node availability, or high churn rates—the test sequence ensures that the anomaly detection system is exposed to diverse edge cases. Performance metrics such as detection latency, false positive rate, and precision-recall are recorded for each scenario. This approach not only validates the system's effectiveness in recognizing anomalous node behaviors but also provides insights into its operational limits and adaptability in realistic adversarial environments.

RUN TIME VISUAL DASHBOARD

- A visual network topology with:
  - **Honest Nodes** (DISPLAYED AS?): Normal behavior — validating, propagating.
  - **Malicious Nodes** (DISPLAYED AS?): Varying behaviors (e.g., Sybil, DoS, eclipse).
  - **Target Node** (highlighted): Observed for isolation or misinformation.
  - **Monitoring Layer** (WHAT CALL?): Detection system running and collecting logs/metrics.
- **Stages** labeled on the diagram:
  - Baseline: All honest.
  - Sybil attack: Flooded with identities.
  - DoS: Message suppression.
  - Eclipse: Target node cut off from honest peers.

The dashboard visually demonstrates how the simulated attacks are introduced and how the detection system is observing them.

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About the SPARK AI Consortium

The SPARK AI Consortium was founded in 2024 as a response to the rapidly evolving AI landscape and the growing need for interdisciplinary collaboration to address the complex challenges of AI implementation in business contexts.

Hosted at the San Diego Supercomputer Center (SDSC), SPARK AI brings together the technical expertise of Center AI researchers in advanced computation, data science, and integrative AI solutions with the expertise, experience, and practical insights of industry leaders to create a unique ecosystem for innovation and thought leadership.

The consortium is built around an eclectic mix of AI experts, senior data professionals, general managers, and academic researchers, united by a common goal: to ensure that AI technologies are developed and deployed in ways that create sustainable value for businesses while benefiting society as a whole.

An advisory board made up of corporate members, research faculty and industry experts guide the consortium in its strategy, operations and publication of research outputs.

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Contact Information

The SPARK AI Consortium
San Diego Supercomputer Center
UC San Diego
9500 Gilman Drive, 0519
La Jolla, CA 9203-0519
Telephone (858) 534-5014
Website: https://sparkai.network/
SDSC: www.sdsc.edu

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